

A devilish question for option sellers: Which VRP is higher?

Kris Abdelmessih (@KrisAbdelmessih) — Moontower · Article on X · Originally published April 2026

A well-supported and common belief in the options world is that there is a premium of implied volatility to realized volatility. The implication is that options, on average, are overpriced, so the structural edge in the options market comes from selling them to “harvest the VRP”.

If you put a gun to my head to confess where I stand, I’ll coldly chant “I have no quarrel with that belief” in the cadence of *His name is Robert Paulson*.

There’s enough evidence to not disagree with the belief. But as a trader who spent 2 decades being long gamma and vega more often than short and not being especially rare among professional option traders in that regard, it’s hard to bang on the VRP desk with fervor.

Aside

It is entirely possible that my trading profits made up for a persistent negative edge in my positioning. That’s plausible because I traded thousands of contracts a day for 20 years in a market-neutral way, but there are facts that cast doubt on the long vol bias being detrimental. My largest p/l’s were in years that were most volatile like the late 2000s, 2018, and 2020, and driven by vega. Now vega is a funny thing because people will argue that it’s not real, it’s marks, etc. They are still on their option blue belt and failing to appreciate

a) that vega can be ported through time through calendars, but it’s almost impossible to do this efficiently unless you are organized to trade flow

b) vega is illiquid. You can manufacture a near-dated option through replication so even if the option is illiquid as long as you can trade an underlying against it there is an out. Deferred options can turn into Hotel California quickly, and when they do it’s good to have the keys to the rooms.

In fact, the market for vega (ie longer-dated options) and gamma (near-dated options) is different. Different players, motivations, and execution methods. For the pros reading, have you ever seen a market where one trader deals in the “fronts” and another trader is in charge of the “backs”. This is the most salient demonstration of this bifurcation, but it exists in subtler ways in many markets and there’s money in bridging the liquidity.

We can peel back assumptions about VRP all day. For example, what do you mean by “realized volatility”? But a highly tiresome unpacking would leave us in a predictably unsatisfying duality:

- Options are generally overpriced
- This is hard to translate into risk-adjusted, opportunity-cost-aware amounts of food on the table

Anyway, I’ve needled you enough. I’m not sorry about that. Part of my self-anointed duty is to test your armor before you march into your Robinhood options tab.

Let’s turn to the “V” in VRP. I’ve seen it refer to “volatility” or “variance”. In the non-technical context that it’s used, this is fine. In our app, it refers to “volatility”. But the distinction is absolutely critical when evaluating and sizing trades.

We can examine this by starting with a market observation that points to expensive options. When we pull it apart, there’s so much more than meets the eye. This lesson brings together fundamentally important themes we cover in this letter like:

- The importance of proper measurement
- how you want to approach the same idea from multiple angles, because it’s rare that a single measure sees all the facets
- The math of option risk and p/l
- Markets being smarter than they seem at first glance

Let’s start with this chart of 30d constant maturity implied vol vs trailing 30d realized vol in HYG. We can see that absent the Liberation Day period last year and the current Iran stress, IV maintains a steady premium to RV. Also note that realized vol is typically below 5% annualized. This is a low-vol ETF.

Another way to look at this is to turn that relationship into a ratio representing the percent premium of IV to RV and call that the volatility risk premium (VRP). We do that below as a scatterplot with the 30-day RV percentiled over the past year. It shows the premium is often

around 40 to 100%, and it's fattest when the realized vol is very low. This is indicative of IV's habit of pricing in mean reversion. When realized vol gets very low, the IV doesn't chase it all the way down. It's stickier as the market doesn't extrapolate the low vol periods to persist. Same when the market gets very volatile. The premium narrows as the market doesn't bid the IV up "too high" as it expects the elevated realized vol to subside.

How you measure the VRP is a choice. We chose to measure it as a ratio because it makes cross-sectional comparison easier. If a name realizes 5% vol but trades for 6% vol, that is put on par with a 50% vol name trading for 60% IV. We could have also chosen to measure the premium as a "spread" instead of a ratio. In that case, HYG could be said to have a VRP of 1 (ie 6% IV - 5% RV) and the higher vol name would have a spread of 10 (60% IV - 50% RV). If HYG was 15% IV and 5% RV, that would also be a 10-point spread, so the spread approach just feels more wrong when you compare assets of different vols. The ratio offers a better comparison.

But remember: how you want to approach the same idea from multiple angles, because it's rare that a single measure sees all the facets. The plot is going to thicken considerably because putting on risk and screening are 2 different activities that are related via decision-making but divorced in execution.

Back to HYG. Whether you use a ratio or spread, implied vol is sitting well above trailing realized vol. The ratio is more flattering than the spread, but either way it appears like a structural risk premium. Sell the options, collect the premium, repeat. Right? As you zoom in, the details start to hint at why this ratio or spread is just sitting there for the taking.

The Gamma-Theta Tug of War

If you are selling options to capture VRP, your intent or thesis is not about IV falling, just that your theta p/l, the option decay you collect as time passes, will outweigh the losses you incur from being short gamma as the stock moves around. We make allowances for path dependence, but on average, if the stock moves around less than what is implied by the option prices, you win.

The net P&L of the position due to realized vol makes the battle of gamma and theta visible. This is from the long option holder's point of view. For them, if the RV is greater than the IV, this term is positive:

$$\text{Net P\&L} \approx \frac{1}{2} \cdot \Gamma \cdot S^2 \cdot (\sigma^2_{\text{RV}} - \sigma^2_{\text{IV}}) \cdot (1/252)$$

Although this comes back to that letter “V”. σ^2 refers to variance not volatility. If you are short options, you make money when realized variance is below implied variance. The distinction between volatility and variance is critical.

We also want to quickly review approximations for ATM options (technically ATF or “at-the-forward”). This is the straddle and its gamma, respectively:

$$\begin{aligned} \text{Straddle} &\approx 0.8 \cdot S \cdot \sigma \cdot \sqrt{T} \\ \Gamma &\approx 0.8 / (S \cdot \sigma \cdot \sqrt{T}) \end{aligned}$$

Focus on the following relationships. First, the straddle price is proportional to σ . Which means daily theta decay is proportional to σ , since you can think of theta as the diff of the straddle formula on T vs the straddle formula on T-1. A 50-vol name bleeds twice as fast as a 25-vol name. Γ is inversely proportional to σ . A 50-vol name has half the gamma per dollar of notional as a 25-vol name. You should notice... a lower vol name has less theta AND more gamma.

Theta-weighting position size

Say you want to sell the same dollar amount of daily theta across two names, a 50-vol name and a 10-vol name. Since theta scales proportionally with vol, you need 5x more contracts in the 10-vol name to collect the same daily premium. That’s the theta-neutral scalar: $\sigma_{\text{high}} / \sigma_{\text{low}}$.

We start with a question: If both names trade at the same ratio premium, say IV is 20% above RV for each, does theta-neutral sizing give you the same variance edge? Holding that ratio premium fixed at 20%, the variance edge ($\text{IV}^2 - \text{RV}^2$) collapses as vol falls. When you scaled notional by 5x in the 10-vol name to match theta, you are capturing much less variance edge for the same risk. To fully equalize variance edge at theta-neutral sizing, you’d need to scale by $(\sigma_{\text{high}} / \sigma_{\text{low}})^2$, the square of the theta-neutral ratio. Theta-neutral means 5x. Variance-edge-neutral means 25x.

How Rich Does The Low-Vol Name Need To Be?

So a 50-vol name trades at 60 vol for a 20% ratio premium. Variance edge per contract: $60^2 - 50^2 = 3600 - 2500 = 1,100$ variance units. You want to find the IV for a 10-vol name such that theta-neutral sizing (5× contracts) delivers the same 1,100 variance units. Working through the algebra, the answer is:

$$IV_{low} = RV_{low} \times \sqrt{1 + (IV_{high}^2 - RV_{high}^2) / (RV_{low} \times RV_{high})}$$

Plugging in: $IV_{low} = 10 \times \sqrt{1 + 1100 / (10 \times 50)} = 10 \times \sqrt{1 + 2.2} = 10 \times \sqrt{3.2} = 17.9$. So the 10-vol name needs to trade at 17.9 vol, a 79% ratio premium to RV, to deliver the same theta-neutral variance edge as the 50-vol name at a 20% premium. A 10-vol name trading at 12 vol (20% premium) looks identical to the 50-vol name on a ratio chart. In variance-edge terms, it's delivering less than a tenth of the expected profit, even after upsizing to theta-neutral scaling.

Spread Gets You Closer

Screening by vol ratio might offer sensible comparisons but fails to compare opportunity in a risk-aware way. At the end of the day, a 40% vol premium on a 5-vol name is still just 2 points. You can get a feel for this when you look at selling options on fixed income ETFs. That IEF vol looks high, and then when you look at the premium it just feels like the juice isn't worth the squeeze.

So what about vol-point spread? It works quite well for comparing edge, at least until you get to much lower vol names. The spread needed for variance-edge equivalence is pretty constant from 25 to 100 vol names. The ratio needed to equate the variance edge went from 20% to 79% to 132%. It makes low-vol names look rich when they're structurally disadvantaged in variance-edge terms.

Back To HYG

The ratio VRP in HYG looks persistently fat, often 50% or more above realized vol. But even the spread measure doesn't quite put it on par with the variance edge in higher vol names. A stock priced at 60 vol that moves at a 50 vol has a 10 point spread, which is equivalent variance edge to HYG moving 5 vol but being priced at 11.6. But that kind of differential is extreme. HYG's median spread in recent data is around 2-3 vol points, not 6 or more,

equivalent to say a name that moves 50 vol trading 55 vol. HYG options really are persistently rich relative to realized vol. But the ratio charts can hide the fact that on a relative basis, HYG vols at a 2-3 point premium are actually cheap compared to a 50 vol name priced at 60% IV!

Additional consideration on low vol names

The cost of harvesting that variance edge premium in low vol names can be more expensive since theta-equivalence means trading far more contracts of the lower vol name (assuming stock prices are the same). And of course, for a given level of theta, the lower vol name has far more gamma, which means more trading costs because you will be trading more shares.

If you are selling options to capture VRP, be aware that the V that matters is variance and that the scaling properties of risk and edge make ranking opportunities a bit unintuitive when looking at ratios. Spreads are a more solid footing, but selling a 20 vol name at 30 vol is still better than selling a 120 vol name at 130 vol, even if ratios can be misleading.

Appendix: Deriving the “Required IV” Formula

I figure out this stuff with paper and pencil moving from one step to another but LLMs are a zillion times better at algebra so I let it derive a formula. I didn't use this formula but it gets you to the same place quickly at the expense of sparing you the intuition-building steps. You provide 3 inputs: RV_low, RV_high, IV_high. It returns the minimum IV the low-vol name needs to trade at for the trade to be economically equivalent on a theta-neutral variance-edge basis. Anything below that number is a relatively worse trade than the baseline you are comparing against.

[I was only able to do the algebra myself until step 6. It was fun to do, like a little puzzle. But then it was depressing to be old and stuck and not even remember quadratic form.]

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